TOTAL OUT OF 30

Last Name: SOLUTIONS

First Name:

Signature:

**University of Waterloo, Mechatronics Engineering**

---

# SYDE252: Linear Systems and Signals
## Dr. Matt Borland
## Midterm Examination
## **Time Allowed: 120 minutes**
## **Rooms: E7-3343, E7-3353**

---

**11:30am - 1:30pm, 19/06/2019**

- *Just you, your brain, an approved calculator, and a pen!*
- *If you choose to write in pencil your test will NOT be eligible for regrading.*
- *Part marks can only be awarded for information that is clearly conveyed*
- *You are strongly advised to take the first 3-5 minutes of the test period to read the entire test over.*
- *An equation sheet has been provided as a separate page.*

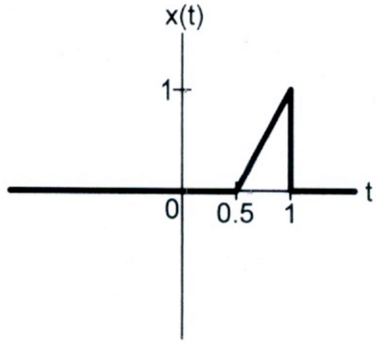
Although this may be a most difficult thing, if one will do it, it can be done. There is nothing that one should suppose cannot be done.

- *Yamamoto Tsunetomo, Hagakure*

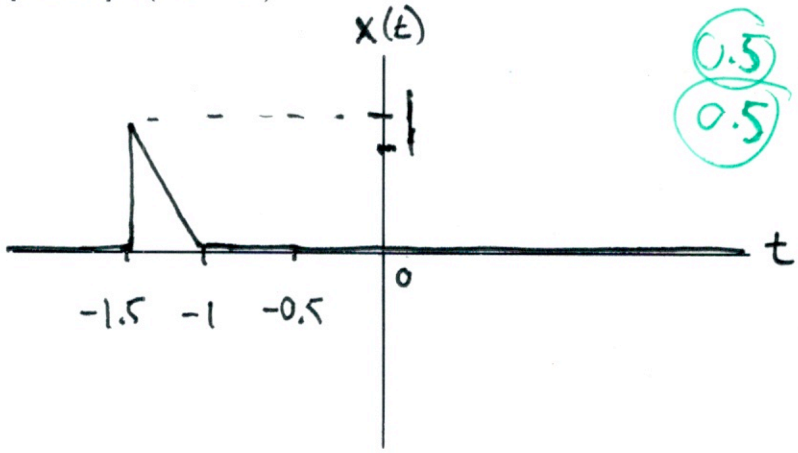
**Do not turn this page until you are told that you may do so.**

2

1. For the signal, $x(t)$, shown below, plot the following transformations. Be as complete as possible:



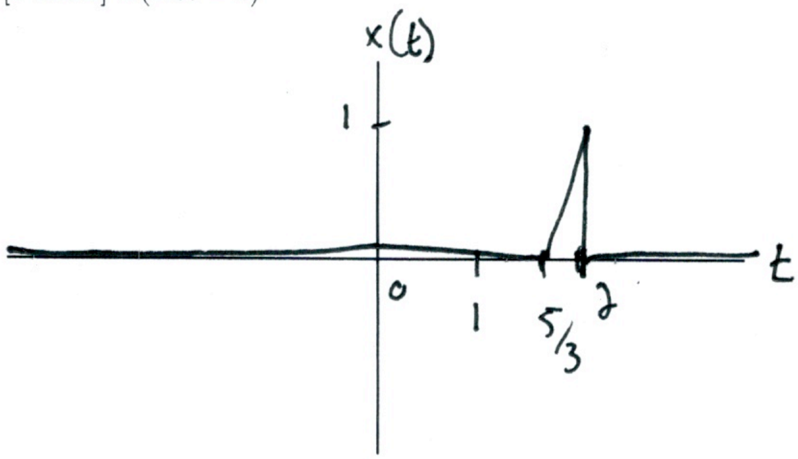

(a) [1 mark] $x(-t-0.5)$




0.5 LABELS IN THE RIGHT SPOTS
0.5 SHAPE/POSITION

- SHIFT 0.5 TO RIGHT
- FLIP

(b) [1 mark] $x(1.5t-2)$



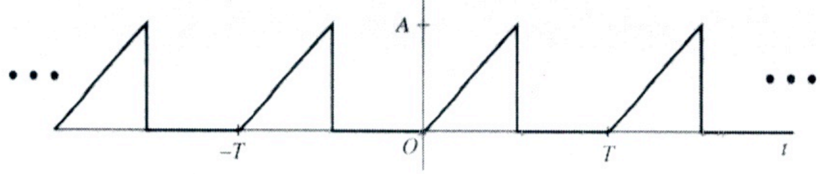

0.5 LABELS IN THE RIGHT SPOT

0.5 SHAPE/POSITION

- SHIFT 2 TO THE RIGHT
- COMPRESS 2/3

3

2. [3 marks] Determine the energy, power, and the RMS value of the 50% duty cycle dc-offset saw-tooth wave $x(t)$ with peak amplitude $A$:




$$x(t) = \begin{cases} \frac{2A}{T} t & 0 \leq t < T/2 \\ 0 & T/2 \leq t < T \end{cases} \qquad 0 \leq t < T$$

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 \, dt \rightarrow \text{SINCE } x(t) \text{ IS PERIODIC}, E_x = \infty$$

0.5 CORRECT ANSWER

$$P_x = \frac{1}{T} \left( \int_0^{T/2} \left(\frac{2A}{T} t\right)^2 dt + \int_{T/2}^{T} 0^2 \, dt \right)$$

$$P_x = \frac{1}{T} \frac{4A^2}{T^2} \int_0^{T/2} t^2 \, dt = \frac{1}{T} \frac{4A^2}{T^2} \left( \frac{t^3}{3} \Big|_0^{T/2} \right)$$

1 METHOD

$$P_x = \frac{1}{T} \frac{4A^2}{T^2} \frac{T^3}{3(8)} = \frac{A^2}{6}$$

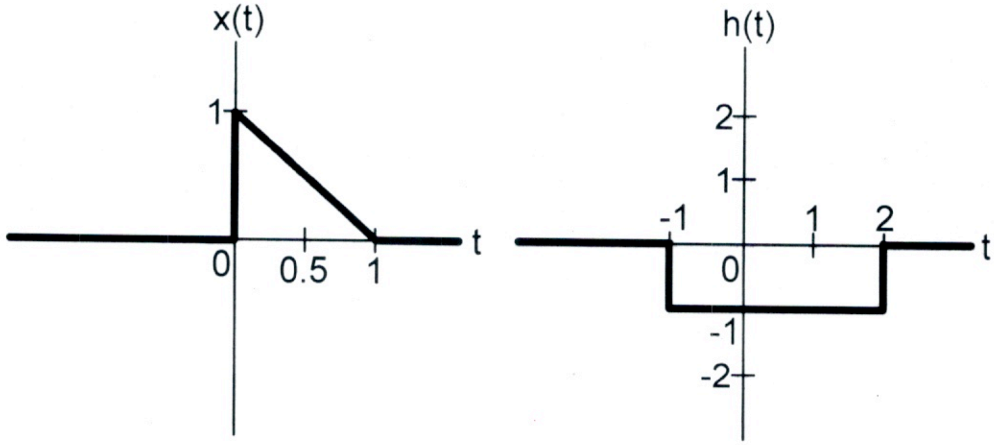
1 CORRECT ANSWER

$$X_{RMS} = \sqrt{P_x} = \sqrt{\frac{A^2}{6}}$$

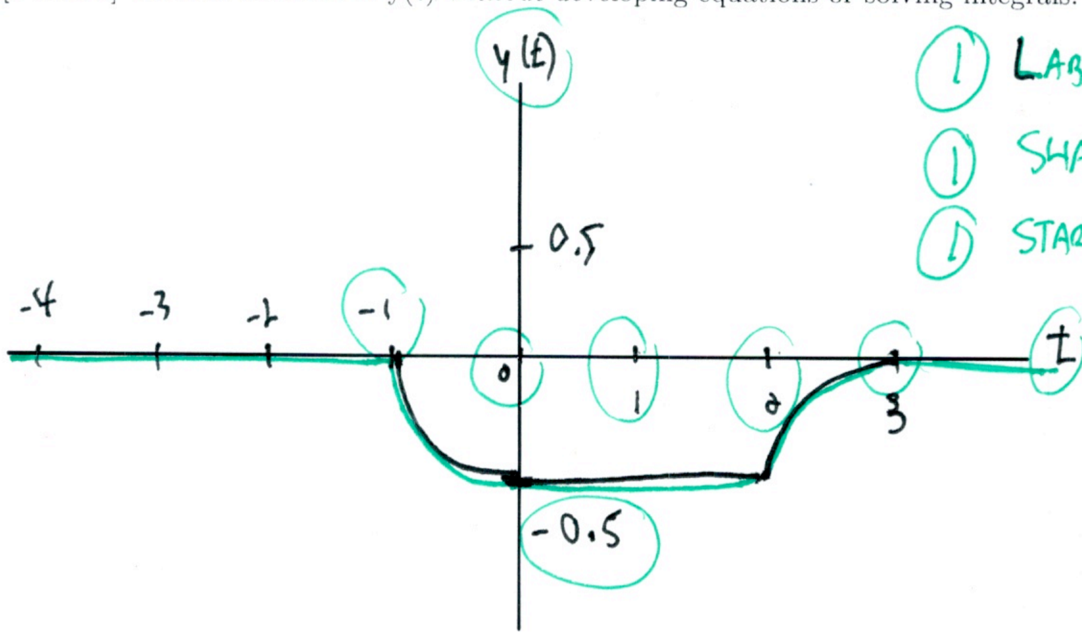
0.5 CORRECT ANSWER.

3

3. For the given signals, x(t) and h(t), use graphical convolution to calculate y(t).




(a) [3 marks] Provide a sketch of y(t) without developing equations or solving integrals:




1 LABELS

1 SHAPE

1 START/STOP TIMES.

7

4. For system defined by:

$(D^2 + 4D + 3)y(t) = (D + 5)x(t)$

(a) [2 marks] Find the Characteristic Equation and solve for its roots.

CHAR EQN: $\lambda^2 + 4\lambda + 3 = 0$

$(\lambda + 1)(\lambda + 3) = 0$ 1 CORRECT EQN

THE CHARACTERISTIC ROOTS ARE

$\lambda_1 = -1$, $\lambda_2 = -3$ 1 CORRECT ANSWER

(b) [1 mark] Find the Characteristic Modes of the system.

CHAR MODES ARE $e^{\lambda t}$

$\therefore$ MODES ARE $e^{-t}$, $e^{-3t}$ 1 CORRECT ANSWER.

...continues on next page...

(c) [4 marks] Find the Unit Impulse Response, $h(t)$, of the system.

USE SIMPLIFIED IMPULSE MATCHING.

$$y_n(t) = C_1 e^{-t} + C_2 e^{-3t} \qquad \text{I.C.'s} \quad y(0) = 0$$

$$\dot{y}_n(t) = -C_1 e^{-t} - 3C_2 e^{-3t} \qquad \dot{y}(0) = 1$$

1 SETUP OF PROBLEM

$$\left.\begin{aligned} 0 &= C_1 + C_2 \\ 1 &= -C_1 - 3C_2 \end{aligned}\right\} \Rightarrow \underbrace{C_1 = 1/2 \qquad C_2 = -1/2}_{0.5}$$

CORRECT ANSWER

CORRECT ANSWER.

0.5 $\rightarrow y_n(t) = \frac{1}{2} e^{-t} - \frac{1}{2} e^{-3t}$

$$h(t) = [P(D)\, y_n(t)]\, u(t) = [(D+5)\, y_n(t)]\, u(t)$$

$$= [\dot{y}_n(t) + 5y_n(t)]\, u(t)$$

1 METHOD

$$= \left[-\tfrac{1}{2}e^{-t} + \tfrac{3}{2}e^{-3t} + \tfrac{5}{2}e^{-t} - \tfrac{5}{2}e^{-3t}\right] u(t)$$

$$= (2e^{-t} - e^{-3t})\, u(t)$$

1 CORRECT ANSWER

7

5. Two inputs, temperature $T(t)$ and wind speed $V(t)$, produce an output, wind chill $W(t)$, according to:

$$W(t) = 35.7 + 0.6\,T(t) - 35.7\,V(t)^{0.16} + T(t)\{V(t)\}^{0.16}$$

...where the independent variable is $t$, time.

For each part below provide a proof or an argument to support your answer.

(a) [1 mark] Is the system BIBO stable?

IF $|T(t)| \le M_T < \infty$ AND $|V(t)| \le M_V < \infty$ 0.5 PROOF

THEN $|W(t)| \le 35.7 + 0.6\,M_T - 35.7\,M_V^{0.16} + M_T M_V^{0.16} < \infty$

THUS A BOUNDED INPUT PRODUCES A BOUNDED OUTPUT,

AND IT IS BIBO STABLE. 0.5 CLEAR ANSWER

NOTE THAT YOU CAN'T HAVE BASICALLY JUST GUESSED AND GOT THE 0.5 MARK.

(b) [1 mark] Is the system memoryless?

0.5 REASON.

CURRENT OUTPUT $W(t)$ DEPENDS ONLY ON CURRENT INPUTS $T(t)$ AND $V(t)$; NO MEMORY IS REQUIRED.

$\therefore$ THE SYSTEM IS MEMORYLESS. 0.5 CLEAR ANSWER

(c) [1 mark] Is the system causal?

→ ALL MEMORYLESS SYSTEMS ARE CAUSAL. 0.5 REASON

OR

→ OUTPUT ONLY DEPENDS ON CURRENT INPUTS

$\therefore$ THE SYSTEM IS CAUSAL. 0.5 CLEAR ANSWER.

...continues on next page...

(d) [2 marks] Is the system linear?

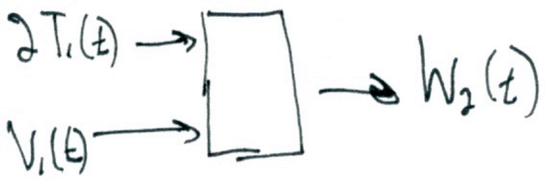

LET $T_1(t) = 35.7$

TEST THE SCALING PROPERTY

$$W_1(t) = 35.7 + 0.6(35.7) - 35.7V(t)^{0.16} + 35.7V(t)^{0.16}$$

$$W_1(t) = 57.12$$

1 APPROPRIATE TEST




$$W_2(t) = 35.7 + 0.6(71.4) - 35.7V(t)^{0.16} + 71.4V(t)^{0.16}$$

$$= 78.54 + 35.7V(t)^{0.16}$$

CHECK SCALING

$$2W_1(t) \neq W_2(t)$$

0.5 CORRECT CONCLUSION OF THE TEST

$\therefore$ THE SYSTEM FAILS SCALING (HOMOGENEITY) AND IS THEREFORE NOT-LINEAR. 0.5 CORRECT ANSWER

$\rightarrow$ COULD HAVE CHECKED ADDITIVITY + HOMOGENEITY TOGETHER...

...continues on next page...

(e) [2 marks] If the temperature is constant, that is $T(t) = k_T$, is the system linear?

$$V_1(t) \longrightarrow \boxed{S} \longrightarrow W_1(t)$$

T(t) IS A CONSTANT $K_T$

TEST SCALING.

$$W_1(t) = 35.7 + 0.6k_T - 35.7\,V(t)^{0.16} + k_T\,V(t)^{0.16}$$

$$V_2(t) = 2V_1(t) \longrightarrow \boxed{S} \longrightarrow W_2(t)$$

1 APPROPRIATE TEST

$$W_2(t) = 35.7 + 0.6k_T - 35.7\left(2V_1(t)\right)^{0.16} + k_T\left(2V_1(t)\right)^{0.16}$$

CHECK SCALING

$$2W_1(t) \neq W_2(t)$$

0.5 CORRECT CONCLUSION

∴ THE SYSTEM FAILS SCALING (HOMOGENEITY) AND IS THEREFORE NON-LINEAR. 0.5 CORRECT ANSWER.

→ COULD HAVE CHECKED ADDITIVITY + HOMOGENEITY TOGETHER...

4

6. Find the zero-state response to the input, x(t), given the system's unit impulse response:

$h(t) = [2e^{-3t} - e^{-2t}]u(t)$

Do not use graphical convolution.

(a) [2 marks] $x(t) = e^{-t}u(t)$

$$y_{ZSR} = x(t) * h(t) = (2e^{-3t} - e^{-2t})u(t) * e^{-t}u(t)$$

$$= 2e^{-3t}u(t) * e^{-t}u(t) - e^{-2t}u(t) * e^{-t}u(t)$$

$$= \left[2\frac{(e^{-t} - e^{-3t})}{2} - \frac{e^{-t} - e^{-2t}}{1}\right]u(t)$$

1 METHOD

$$y_{ZSR} = (e^{-2t} - e^{-3t})u(t)$$

1 CORRECT ANSWER

* NOTE I SWAPPED THE ORDER TO AVOID DOUBLE NEG.

FROM TABLE 2.1

#4

$$e^{\lambda_1 t}u(t) * e^{\lambda_2 t}u(t) = \frac{e^{\lambda_1 t} - e^{\lambda_2 t}}{\lambda_1 - \lambda_2}u(t)$$

(b) [2 marks] $x(t) = e^{-2t}u(t)$

$$y_{ZSR} = x(t) * h(t) = (2e^{-3t} - e^{-2t})u(t) * e^{-2t}u(t)$$

$$= 2e^{-3t}u(t) * e^{-2t}u(t) - e^{-2t}u(t) * e^{-2t}u(t)$$

$$= \left[2\frac{(e^{-2t} - e^{-3t})}{1} - te^{-2t}\right]u(t)$$

1 METHOD

$$= \left[(2-t)e^{-2t} - 2e^{-3t}\right]u(t)$$

1 CORRECT ANSWER

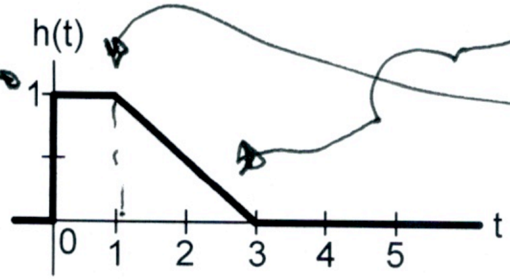
FROM TABLE 2.1

#4 & #5

#5 $$e^{\lambda t}u(t) * e^{\lambda t}u(t) = te^{\lambda t}u(t)$$

4

7. A lowpass LTIC system has an impulse response, $h(t)$, shown below.




OR BY INSPECTION

(a) [2 marks] Determine a rectangular approximation of $\hat{h}(t)$ to $h(t)$

$\hat{h}(t)$ NEEDS THE SAME AREA AND HEIGHT OF $h(t)$

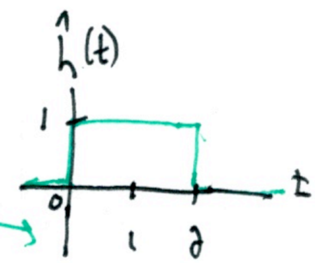
METHOD 1

$$A_h = \int_{-\infty}^{\infty} h(t)\,dt = \int_0^1 1\,dt + \int_1^3 \frac{3-t}{2}\,dt = 1 + \left(\left(\frac{3t}{2} - \frac{t^2}{4}\right)\Big|_1^3\right) = 1 + 1 = 2$$

$$T_h = \frac{A_h}{P_h} = \frac{2}{1} = 2$$

$$\therefore \hat{h}(t) = u(t) - u(t-2)$$

OR GRAPHICALLY 1




(b) [1 mark] Determine the time constant, $T_h$, for $\hat{h}(t)$.

$$T_h = \frac{A_h}{P_h} = \frac{2}{1} = 2\,s$$ 0.5 CORRECT ANSWER

0.5 CORRECT RELATIONSHIP

(c) [½ mark] Using $\hat{h}(t)$, what is the approximate radian cutoff frequency, $\omega_c$, of this system?

$$f_c = \frac{1}{T_h} = \frac{1}{2} = 0.5\,Hz \qquad \omega_c = 2\pi f_c = \pi \text{ rad/s}$$ 0.5

(d) [½ mark] Assuming a frequency $\omega_0 >> \omega_c$, what is the system response, $y(t)$, to the input $x(t) = \sin(\omega_0 t + 3.674\pi)$. Provide a numerical answer.

$$y(t) = 0$$ 0.5

THE SYSTEM IS A LOW PASS FILTER, SO THE $\omega_0 >> \omega_c$ IS BLOCKED.

**EXTRA PAGE FOR ROUGH WORK -** Nothing on this page will be graded. Answer everything in the space provided in the questions.